04_composite_indexes_column_order

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

PURPOSE

- Demonstrate how column order in a composite index
- Determines whether the optimizer can perform a Seek or is forced into a Scan. The B-tree is sorted left to right. The first key
- Column defines the primary sort. A query that filters only on the second column cannot navigate the B-tree efficiently.

4.0 ENVIRONMENT CHECK

```
USE STN_Lab;
GO

-- ============================================================
-- 4.0 -- ENVIRONMENT CHECK
-- ============================================================

SELECT
    i.index_id,
    i.name,
    i.type_desc,
    STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal) AS key_columns
FROM sys.indexes i
JOIN sys.index_columns ic
    ON i.object_id = ic.object_id
   AND i.index_id  = ic.index_id
JOIN sys.columns c
    ON ic.object_id = c.object_id
   AND ic.column_id = c.column_id
WHERE i.object_id = OBJECT_ID('dbo.Declaracion')
  AND ic.is_included_column = 0
GROUP BY i.index_id, i.name, i.type_desc
```

```
ORDER BY i.index_id;
GO
```

4.1 COMPOSITE INDEX: HIGH SELECTIVITY FIRST

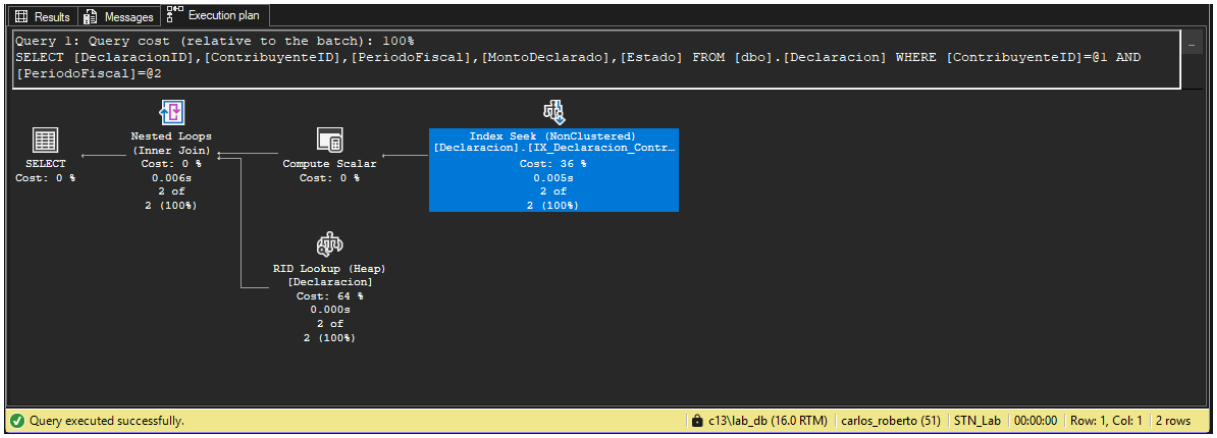
Orden: ContribuyenteID (alta selectividad), PeriodoFiscal Query filtra por ambas columnas.
El optimizador navega a ContribuyenteID primero, luego reduce por PeriodoFiscal dentro de esa particion.
Expected: Index Seek en ambos predicados

```
USE STN_Lab;
GO

SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE ContribuyenteID = 121644
  AND PeriodoFiscal = '2023-04';    -- replace with existing period

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



OBSERVACION 4.1

- reads_4.1 = 5
- Operator = Index Seek (ambos predicados usados: ContribuyenteID + PeriodoFiscal)
- RID Lookup presente por DeclaracionID fuera del INCLUDE - costo 64%

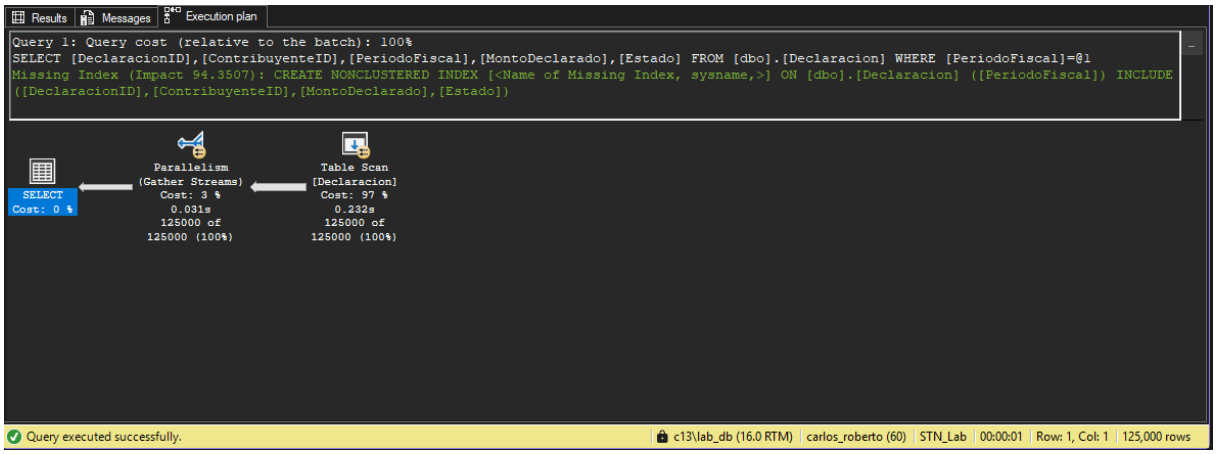
4.2 MISMO INDICE, FILTRO SOLO EN SEGUNDA COLUMNA

```
USE STN_Lab;
GO

SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE
    PeriodoFiscal = '2023-04';    -- primera columna ausente del filtro


SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



OBSERVACION 4.2

- reads_4.2 = 13852
- Operator = Index Scan (segunda columna sola no puede hacer Seek)

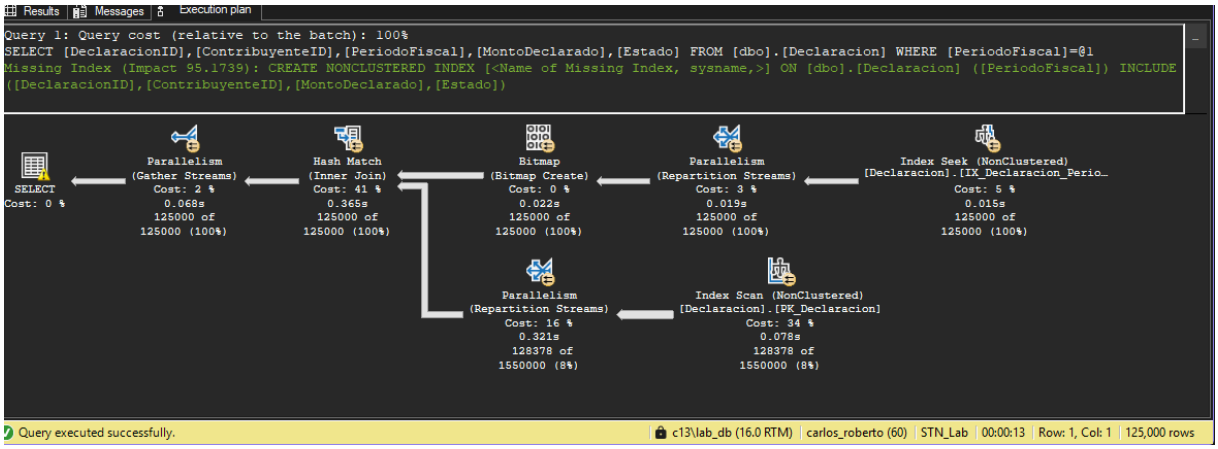
4.3 ORDEN INVERTIDO: PeriodoFiscal PRIMERO

```
--crear el indice
CREATE NONCLUSTERED INDEX IX_Declaracion_Periodo_Contribuyente
    ON dbo.Declaracion (PeriodoFiscal, ContribuyenteID)
    INCLUDE (MontoDeclarado, Estado);
GO

SET STATISTICS IO ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    PeriodoFiscal,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE PeriodoFiscal = '2023';

SET STATISTICS IO OFF;
GO
```



4.4 IGUALDAD vs RANGO: EL ORDEN IMPORTA MAS AQUI

Predicados de igualdad antes que predicados de rango. Un predicado de rango (BETWEEN, >, <) detiene el B-tree de usar columnas clave subsiguientes para el Seek.

4.4.1 Orden correcto: igualdad primero, rango segundo

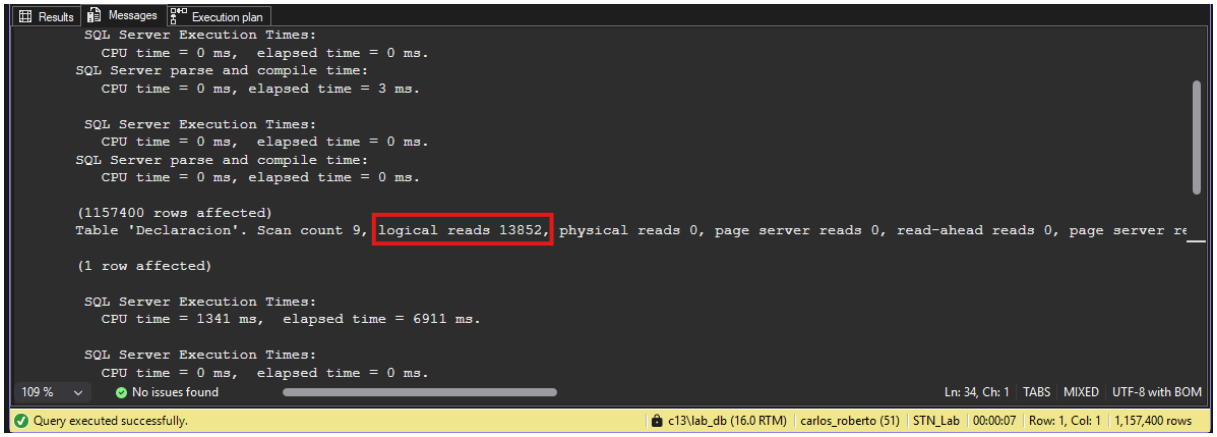
```
USE STN_Lab;
GO

--CREAR INDICE
CREATE NONCLUSTERED INDEX IX_Declaracion_Estado_Fecha
    ON dbo.Declaracion (Estado, FechaPresentacion)
    INCLUDE (ContribuyenteID, MontoDeclarado);
GO

SET STATISTICS IO ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    FechaPresentacion,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE Estado = 'PR'
  AND FechaPresentacion >= '2020-01-03';

SET STATISTICS IO OFF;
GO
```



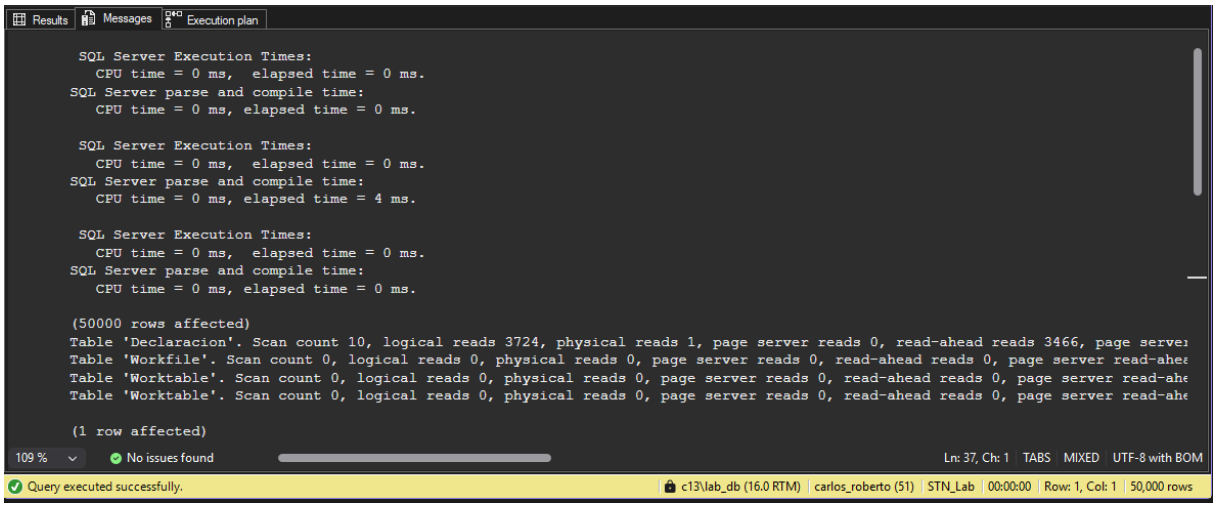
4.4.2 Orden incorrecto: rango primero, igualdad segundo

```sql
CREATE NONCLUSTERED INDEX IX_Declaracion_Fecha_Estado
    ON dbo.Declaracion (FechaPresentacion, Estado)
    INCLUDE (ContribuyenteID, MontoDeclarado);
GO

SET STATISTICS IO ON;

SELECT
    DeclaracionID,
    ContribuyenteID,
    FechaPresentacion,
    MontoDeclarado,
    Estado
FROM dbo.Declaracion
WHERE Estado = 'P'
  AND FechaPresentacion >= '2023-01-01';

SET STATISTICS IO OFF;
GO
```



4.5 COMPARACION DE TAMANO ENTRE VERSIONES DEL INDICE

```sql
SELECT
    i.name,
    s.used_page_count * 8   AS size_kb,
    s.row_count
```

```
FROM sys.dm_db_partition_stats s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
WHERE s.object_id = OBJECT_ID('dbo.Declaracion')
  AND i.name IN (
    'IX_Declaracion_Contribuyente_Periodo',
    'IX_Declaracion_Periodo_Contribuyente',
    'IX_Declaracion_Estado_Fecha',
    'IX_Declaracion_Fecha_Estado'
)
ORDER BY i.name;
GO
```

Results | Messages | Execution plan

	Nombre	size_kb	row_count
1	IX_Declaracion_Contribuyente_Periodo	55640	1550000
2	IX_Declaracion_Estado_Fecha	55560	1550000
3	IX_Declaracion_Fecha_Estado	55672	1550000
4	IX_Declaracion_Periodo_Contribuyente	55696	1550000

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (51) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 4 rows

4.6 CLEANUP

```
DROP INDEX IF EXISTS IX_Declaracion_Contribuyente_Periodo ON dbo.Declara
cion;
DROP INDEX IF EXISTS IX_Declaracion_Periodo_Contribuyente ON dbo.Declara
cion;
DROP INDEX IF EXISTS IX_Declaracion_Estado_Fecha          ON dbo.Declara
cion;
DROP INDEX IF EXISTS IX_Declaracion_Fecha_Estado          ON dbo.Declara
cion;
```